



Jean-Francois Panisset

Technology Committee
Visual Effects Society

Nick Cannon

Technology Committee
Visual Effects Society

**Introducing the Wayland
for Artists ASWF Working
Group**

2:00-2:10 PM



AlmaLinux Day: LA



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Wayland for Artists Working Group

Who are we?

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- VES Technology Committee
 - VFX Reference Platform
 - A per year consistent build environment and components for commercial and in house software
 - Alignment between DCC vendors, studios and open source projects
 - <https://vfxplatform.com>
- Academy Software Foundation
 - A home for critical VFX/Animation projects
 - OpenColorIO
 - OpenEXR
 - ... so many more
 - Also an infrastructure for collaboration: Working Groups

Component	CY2027 DRAFT	CY2026	CY2025	CY2024	CY2023
• Linux					
gcc ↓↑	14.2	14.2	11.2.1 (New libstdc++ ABI) (see notes)	11.2.1 (New libstdc++ ABI) (see notes)	11.2.1 (Switch to new libstdc++ ABI) (see notes)
glibc ↓↑	2.34	2.28	2.28	2.28	2.28
• macOS					
Minimum Deployment Target	15.0 (see notes)	14.0 (see notes)	12.0 (see notes)	11.0 (see notes)	11.0 (see notes)
• Windows					
Minimum Platform Toolset	Visual Studio build tools v14.44 or later (see notes)	Visual Studio 2022 v17.6 or later	Visual Studio 2022 v17.6 or later	Visual Studio 2022 v17.4 or later	Visual Studio 2022
Windows SDK	10.0.22621 or later	10.0.22621 or later	10.0.20348 or later	10.0.20348 or later	10.0.19041 or later
• Common Components					
C++ API/SDK	C++20	C++20	C++17	C++17	C++17
Python	3.13.x	3.13.x	3.11.x	3.11.x	3.10.x
PyQt	6.8.x	6.8.x	6.5.x	6.5.x	5.15
Qt for Python (PySide)	6.8.x	6.8.x	6.5.x	6.5.x	5.15
Qt	6.8.x	6.8.x	6.5.x (6.8 planned for CY2026)	6.5.x	5.15.x (6.5 planned for CY2024)
ACES	2.0	2.0	2.0	1.3	1.3
Alembic	1.8.x	1.8.x	1.8.x	1.8.x	1.8.x
Boost	1.91	1.88	1.85	1.82	1.80
FBX	2020.2 - 2020.latest	2020.2 - 2020.latest	2020.2 - 2020.latest	2020.2 - 2020.latest	2020.2 - 2020.latest

Linux Workstations in M&E: A Brief History

When dinosaurs roamed the earth: SGI and IRIX

- CPU, system, graphics, audio, video, OS, compilers, desktop from same vendor
- IrisGL: window management / input handling bundled with 2D / 3D API
- Also X/Motif, or simpler toolkits like Forms / FLTK
- In-house apps at first, then commercial apps
- Basically no open source, except for libtiff, libjpeg: you had to write it all yourself
- So many Perl scripts...

Natural transition to Linux / x86 around 2000

- OpenGL integrates with X11 (GLX)
- Python replaces Perl
- Qt takes over from custom UIs
- No real consensus on desktop environment: GNOME, KDE, others
- Vulkan starting to displace OpenGL
- Cross platform technologies usually preferred
 - macOS and Windows are a must

Industry Transition Challenges

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Python 2 to 3 Transition

- SIGGRAPH BoFs
- Python 3 Dependency Tracker
- VFX Reference Platform CY2020 : Python 3.7.x

CentOS 7 EOL

- Studio Linux Surveys
- 2024 VFX/Animation Studio Workstation Linux Report
 - Recommended transition away from CentOS 7
 - Build on EL 8, but deploy on EL 9

X11 to Wayland

- X11 / Xorg going away in most distributions
- Studios still mostly on EL 8 / EL 9

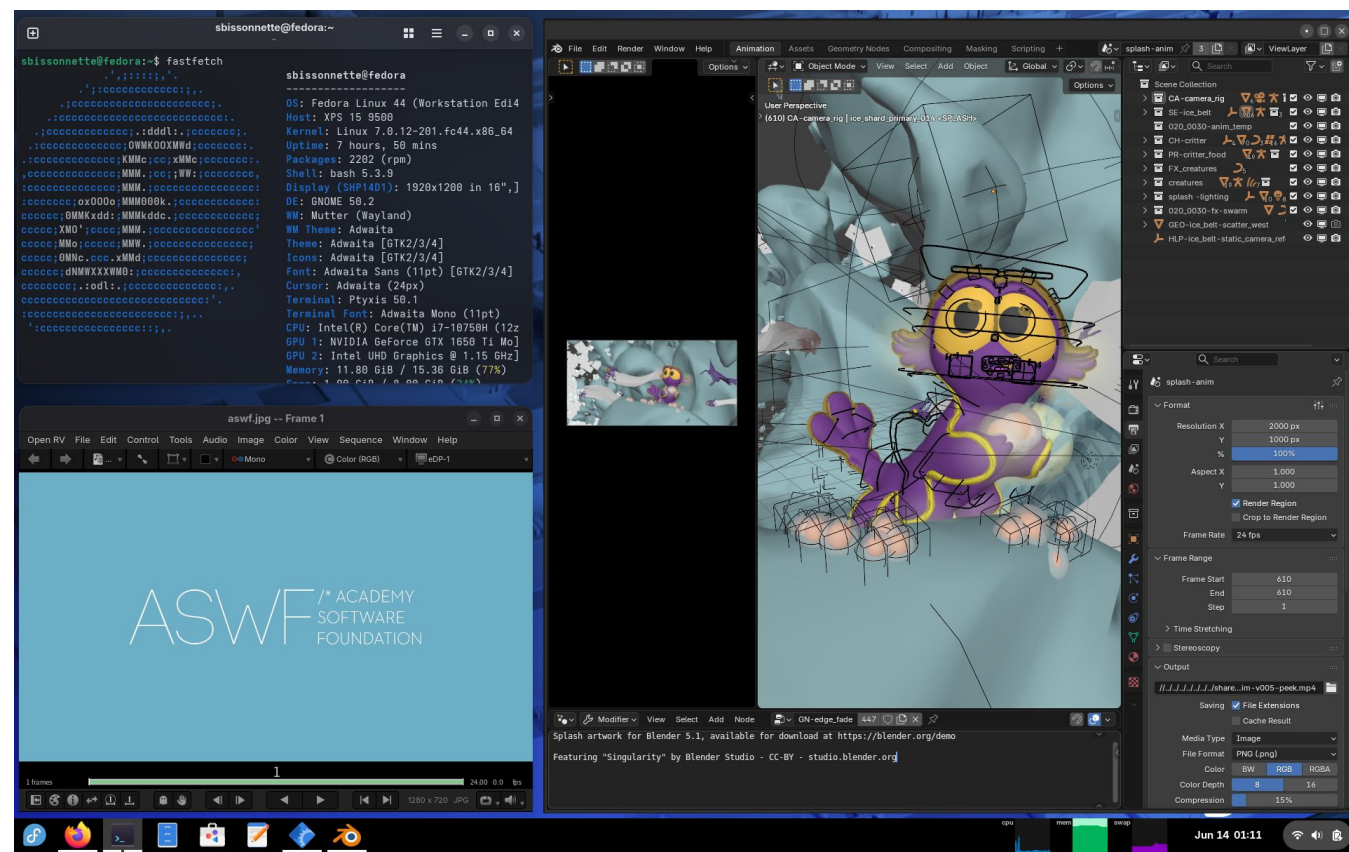


Transition to Wayland Challenges

Linux distributions are dropping support for the Xorg server and moving to Wayland... yet most studios, open source projects, commercial software vendors haven't really investigated the impact of this transition on the Animation / VFX Linux environment.

Specific requirements:

- Conservative version adoption (multi year film projects can't easily change mid schedule)
- Graphics tablet support
- Color management / HDR support
- Legacy software compatibility (in house and commercial)
- Multi monitor support
- NVIDIA GPU support
- High performance remote access
- ...



Wayland for Artists ASWF Working Group

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The Wayland for Artists Working Group will explore a range of industry-specific areas that may be impacted, including graphics tablet support, window management, input focus, key bindings, desktop customization, remote access solutions, GPU acceleration, color management, High Dynamic Range displays, and compatibility with commercial, open source, and in-house applications. Ultimately, the Working Group plans to align stakeholders on a set of shared goals, with a clear roadmap for implementation.

We aim to provide a vendor neutral forum to explore issues related to this transition, gather and organize information that everyone can use. We are not trying to port applications or mandate the adoption of a specific desktop environment. We want to find out what "just works" (XWayland), what needs small fixes, what requires a full port...

Working Group Resources

A Wiki to aggregate information: <https://lf-aswf.atlassian.net/wiki/spaces/WFA/overview>

A monthly Zoom call open to all

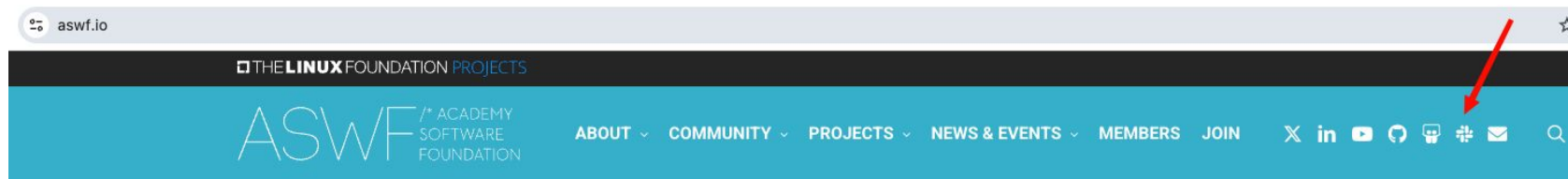
- 2 so far, 20-30 participants, lively discussion and open exchange of information
 - Discussion on Wayland support in Qt versions helping to shape future VFX Platform versions
 - Interest around prototyping high performance encoding for remote access
- Next call would normally be on Tue 2026-07-21 at 13:00 PDT but will push to 2026-07-28 due to SIGGRAPH (normal cadence to resume in August)
- Join via ASWF Calendar: <https://www.aswf.io/meeting-calendar/>
- Meeting notes posted to Wiki

#wg-wayland channel on the ASWF Slack

Call to Action

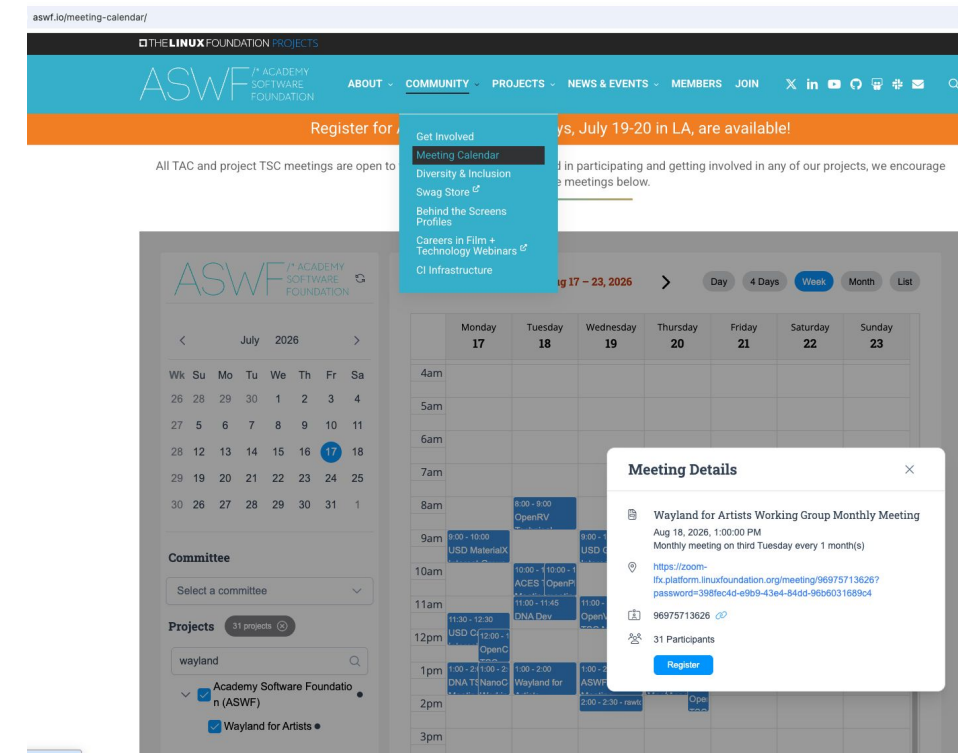
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Join us on Slack: go to <https://aswf.io> and click on the Slack icon to join, you will probably find several other channels of interest



Community -> Meeting Calendar, look for the "Wayland for Artists" meetings and Register

Show us what you've already done with Wayland, we need to learn together



Q&A